

12.3-12.5

Multielectron atom and the periodic table

Electrons shield the nucleus. We have pairwise coulomb interactions between each electron + electrons to the nucleus

$$V(r) = \begin{cases} -\frac{Ze^2}{r} & r \rightarrow 0 \\ -\frac{e^2}{r} & r \rightarrow \infty \end{cases}$$

The orbital shells and their energies!

12.4 Covalent Bonding

$$\langle 1|2 \rangle = \int d^3r \langle 1|\mathbf{r} \rangle \langle \mathbf{r}|2 \rangle \quad (12.69)$$

is called the **overlap integral**. Note that the nonvanishing of the off-diagonal matrix element depends on the states $|1\rangle$ and $|2\rangle$ overlapping in space. Since the wave functions are real, the off-diagonal matrix elements are equal:

$$\begin{aligned} H_{21} &= \langle 2| \left(\frac{\hat{\mathbf{p}}^2}{2m_e} - \frac{e^2}{|\hat{\mathbf{r}} - \mathbf{R}/2|} \right) |1\rangle - \langle 2| \frac{e^2}{|\hat{\mathbf{r}} + \mathbf{R}/2|} |1\rangle + \frac{e^2}{R} \langle 2|1\rangle \\ &= \left(E_1 + \frac{e^2}{R} \right) \langle 2|1\rangle - \int d^3r \langle 2|\mathbf{r} \rangle \frac{e^2}{|\mathbf{r} + \mathbf{R}/2|} \langle \mathbf{r}|1\rangle = H_{12} \end{aligned} \quad (12.70)$$

we underestimate the bonding energy

replacing electrons with muons makes atomic radii smaller, so fusion can happen easier with muonic molecules!?!??

12.5

fermions don't stack, but lets see how this all changes iwth bosons.